Astana IT University
School of Computer Engineering · Software Engineering

DIPLOMA PROJECT · PROBATION PERIOD 1 REPORT

Development of a Real-time Pedestrian Risk Assessment System for Truck Blind Spots Based on Object Detection Methods

Working name of the system: **public.safe** — smartphones as camera nodes, YOLOv8n person detection, foot-point zone-membership rules, live web dashboard

Student	**Bolatov Bekdaulet Almasovich**, group SE-2405 (individual project)
Supervisor	Nurgaliyeva Symbat Altybayevna, PhD, Assistant Professor
Repository	github.com/BekdauletBol/public_final
Submitted version	tag `prob-period-1` — github.com/BekdauletBol/public_final/tree/prob-period-1
Task board	github.com/users/BekdauletBol/projects/1
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All materials in this report correspond to the repository version tagged `prob-period-1`. Editable diagram sources are in `docs/diagrams/*.puml` (PlantUML) with PNG/SVG exports in `docs/diagrams/export/`; this report is stored as `docs/report/SE-2405_Bolatov_ProbPeriod1.pdf`.

1. Project proposal

Project title	Development of a Real-time Pedestrian Risk Assessment System for Truck Blind Spots Based on Object Detection Methods (system name: **public.safe**)
Team	Bolatov Bekdaulet Almasovich, SE-2405 — individual project; sole developer (system design, backend, computer-vision pipeline, frontend, testing, documentation). GitHub: BekdauletBol
Supervisor	Nurgaliyeva Symbat Altybayevna, PhD, Assistant Professor, School of Computer Engineering
Type	Software system with an experimental (research) component; continuation of the student research work "Edge-distributed pedestrian monitoring system using YOLOv8 and geometric spatial analysis for real-time urban danger zone detection" (AITU, 2026)

1.1 Problem statement

Pedestrians account for about 23 % of road deaths worldwide [1], and in Kazakhstan collisions at crossings, loading areas and along the kerb remain frequent because drivers notice pedestrians too late [2]. Large vehicles make the problem worse: the high cabin of a truck or bus creates blind spots at the nearside and directly in front of the vehicle, so a person standing in a loading bay, next to a parked lorry or on the kerb beside a turning truck is often invisible to the driver until the last moment [16]. In the survey conducted for this project (Section 2) 61 % of respondents named high truck/bus cabins as one of the main reasons why drivers fail to notice pedestrians, and 92 % had experienced a near-miss at a crossing.

Existing counter-measures are either passive (lighting, LED markings, signage — they improve visibility but do not analyse the scene) or expensive and slow to deploy (fixed CCTV with analytics servers, estimated at $15 000–50 000 per site in Kazakhstan [3]; retrofit on-vehicle sensor kits that protect one vehicle only). There is no cheap, quickly deployable way to monitor a specific dangerous area — a loading bay, a yard gate, a construction-site exit, the kerb at a crossing — and to raise an alert the moment a person steps into it.

1.2 Target users

- **Safety staff** of logistics yards, warehouses, construction sites and bus depots who need to know when a person enters an area where trucks manoeuvre and cannot see them.
- **Municipal / road-safety researchers** who want to measure how often pedestrians enter a risk zone at a given crossing before deciding on infrastructure changes.
- **Developers and students** who need a reproducible, open pipeline (same code live and offline) for evaluating zone-membership rules on public datasets.

1.3 Aim

To develop and evaluate a low-cost real-time system that turns ordinary smartphones into camera nodes, detects pedestrians with an object-detection network, decides whether each person is inside a configured danger zone using a foot-point geometric rule, and delivers entry/exit events to a web dashboard (and, in future, to a driver-facing warning device) with sub-second latency.

1.4 Objectives

1. Analyse existing pedestrian-safety and video-analytics solutions and formulate requirements grounded in user evidence (survey of pedestrians and drivers).
2. Design a three-tier architecture: phone browser (capture) → laptop backend (FastAPI, YOLOv8n, geometric rules) → web dashboard; record every key decision as an ADR.

3. Implement secure phone pairing (PIN/QR, opaque device tokens, HTTPS/WSS), a per-camera latest-frame scheduler, person detection, six zone-membership rules computed every frame, a hysteresis state machine producing `zone_entered` / `zone_cleared` events, and a dashboard with a polygon zone editor, live overlays, events and CSV export.
4. Build an evaluation harness that runs the same detector and rules offline on public datasets (MOT17, WILDTRACK) and on own recordings, with pre-registered metrics and success criteria.
5. Measure precision/recall/F1 of foot-point rules against the bounding-box-overlap baseline, and tail latency of the latest-frame scheduler against a FIFO queue with several phones.
6. Document, test (unit, integration, security, CV, end-to-end, load) and demonstrate the system; prepare the approbation report.

1.5 Expected outcomes

- A working local system (one command to set up, one to run) with 218+ automated tests, documented in the repository.
- Quantitative answer to the research question: whether foot-anchored zone membership reduces false alarms compared with bounding-box overlap, and at what cost in recall (target: precision ≥ baseline + 0.10 with recall loss ≤ 0.05).
- Measured latency profile: server-side p50 ≤ 150 ms and p99 ≤ 400 ms with three phones at 10 fps on a laptop; fairness ratio ≤ 1.3 between cameras.
- Reusable architecture decisions (9 ADRs), diagrams and an evaluation protocol that other teams can repeat.

1.6 Scope

In scope (diploma)	Out of scope / future work
• Browser-based phone capture (iOS Safari, Android Chrome), PIN/QR pairing, auto-reconnect • Single-laptop backend on a local network; SQLite by default, PostgreSQL optional • YOLOv8n person detection (MPS / CUDA / CPU), six membership rules, hysteresis events • Dashboard: sessions, cameras, zone editor, live overlays, events, metrics, security log • Offline evaluation CLI, load test, documentation, ADRs • Pre-registered experiments E1–E7 (Section 9)	• Native mobile apps; WebRTC; cloud deployment; microservices; message brokers • Person tracking across frames (planned extension behind the detector seam) • Ground-plane calibration / homography; weather-adaptive thresholds • Face recognition or any identity inference (explicitly excluded for privacy) • Physical driver-warning hardware (LED, ESP32 nodes) — concept only • Multi-tenant role model (single admin is sufficient)

2. Evidence of relevance

The requirements are grounded in a structured online survey of pedestrians and drivers (n = 36), which exceeds the 3–5 user discussions expected at this stage, and in WHO road-safety statistics [1, 2]. Findings that are directly measured are separated from assumptions in Section 2.3; interviews at a pilot site (yard or depot) are planned for the next period.

2.1 Survey of pedestrians and drivers (n = 36)

A Google Forms questionnaire with eight closed questions and one open question was distributed digitally among residents of Astana (students, colleagues, acquaintances; convenience sample). It covered (1) pedestrian experience at unsignalised crossings, (2) driver experience and (3) attitude towards automatic warning systems. 36 people answered the pedestrian section; the driver-specific questions were answered by 10 respondents. Raw results are available in the form's response summary (screenshots in Appendix A).

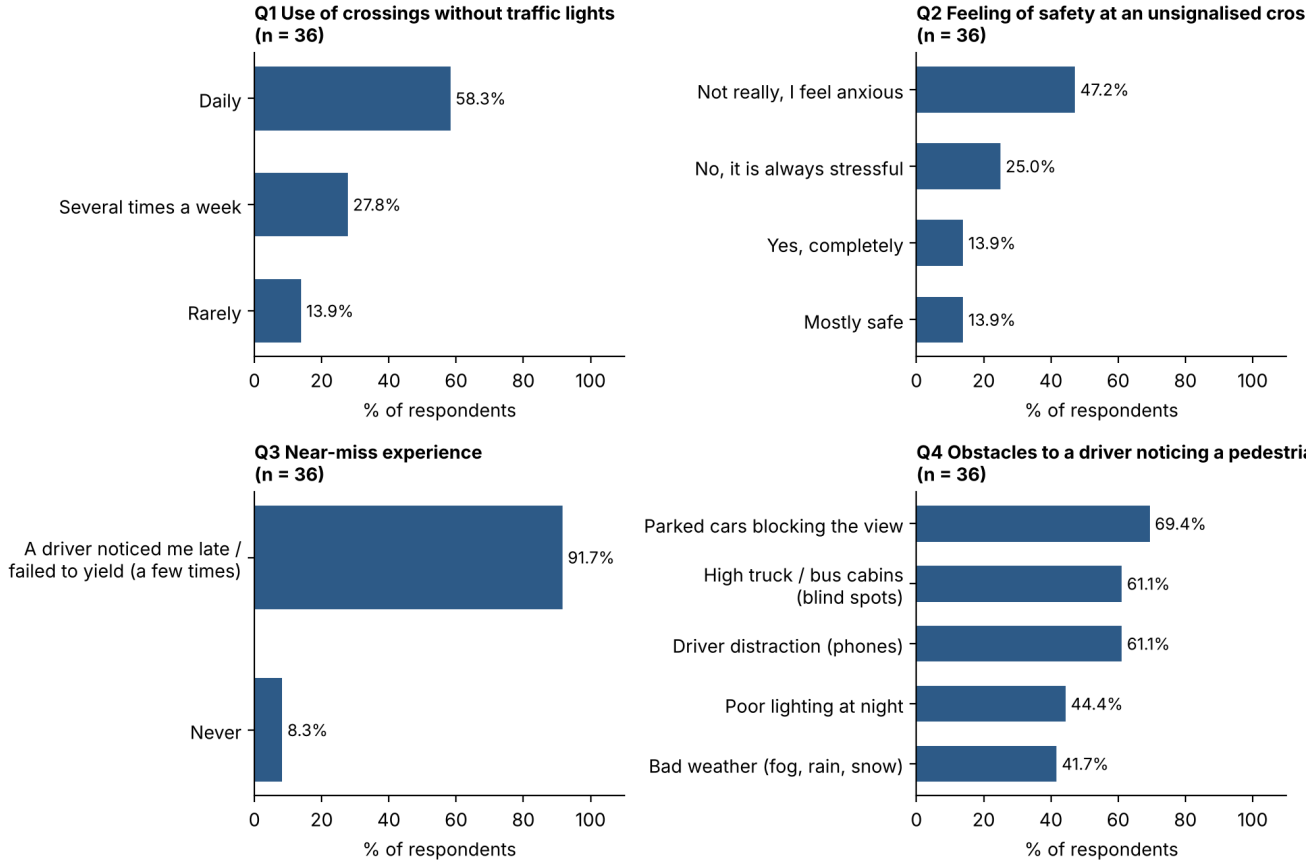


Figure 2.1. Pedestrian section of the survey (n = 36).

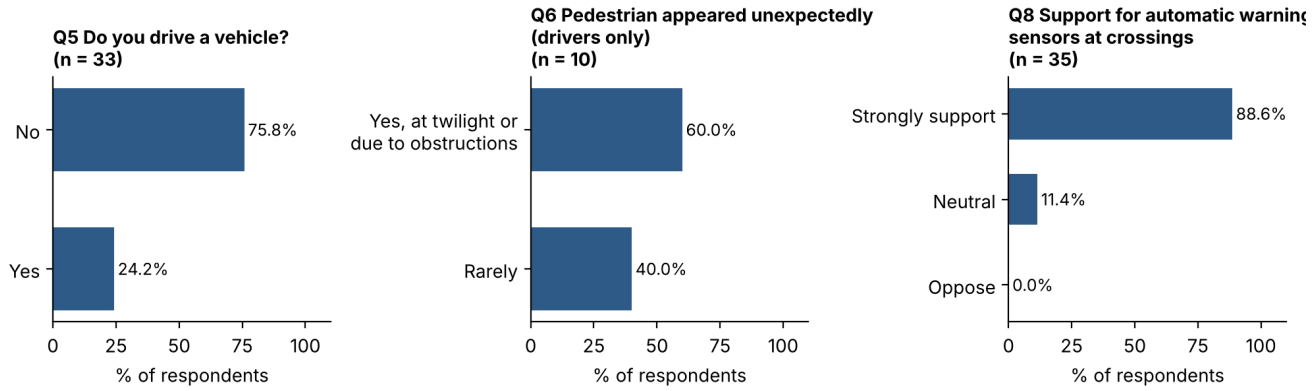


Figure 2.2. Driver section and attitude towards automatic warning (n = 33 / 10 / 35).

Key findings. 86 % of respondents cross unsignalised crossings daily or several times a week; 72 % feel anxious or stressed there; 91.7 % have experienced a driver noticing them late or failing to yield. The most cited visibility obstacles are parked cars blocking the view (69.4 %), **high truck/bus cabins — blind spots (61.1 %)** and driver distraction (61.1 %). 88.6 % strongly support automatic sensors that trigger a warning when a person appears; nobody opposed.

By segment. Among the 10 respondents who answered the driver section, 60 % said that a pedestrian had appeared unexpectedly at a crossing, mostly at twilight or because of obstructions, and all drivers answering Q7 said a blinking indicator warning of a pedestrian entering the crossing would be useful. Pedestrians (the majority of the sample, 75.8 % do not drive) are the group reporting anxiety: 72 % of them do not feel safe at an unsignalised crossing. Both groups therefore describe the same situation from opposite sides — the pedestrian is not seen, and the driver sees too late — and both accept a camera-triggered warning as the remedy.

Open answers (11 responses) repeat three themes — a visible signal for drivers, better lighting, and enforcement — and two of them describe the proposed system almost literally:

- Driver: *"As a driver, I believe safety at crossings can be improved by using smart camera systems that detect pedestrians in real time and automatically alert drivers. For example, an AI-based vision model could recognize when someone is about to cross and trigger warning signals or lights."*
- Pedestrian: *"As a pedestrian, I feel safer when smart cameras detect me and alert drivers as I approach a crossing. This gives me more confidence that drivers will notice and slow down."*
- Others: *"Put the visible signals for drivers to notice people. Make more light at crossings."*; *"visibility should be prioritized by installing high-intensity lighting, reflective signage"*; *"Connect with government and give them penalty"*.

What the survey changed in the requirements. (1) The danger zone must be a free polygon drawn by the operator, because the dangerous area differs between a crossing, a parked-truck nearside and a loading bay (Q4). (2) The system must produce a discrete, debounced event (`zone_entered`) that can drive an external light — a flicker of per-frame detections would be useless for a driver (Q7, open answers). (3) No video is stored and a privacy mode exists, because acceptance of cameras in the open answers is tied to their purpose (warning), not to surveillance. (4) Night operation is a first-class test condition (poor lighting was cited by 44 %; experiment E7).

Limitations: convenience sample, self-reported, 36 respondents; results are indicative and are used to prioritise requirements, not as population statistics.

2.2 Statistical context

- WHO Global Status Report on Road Safety 2023: pedestrians are 23 % of road-traffic deaths worldwide [1].

- Kazakhstan country profile (WHO, 2023): road-traffic deaths remain high; vulnerable road users are a significant share [2]. **[optional: add national statistics from Qazstat / MVD for 2021–2025 if available]**
- Fixed CCTV with analytics costs an estimated $15 000–50 000 per intersection in Kazakhstan and cannot be deployed temporarily [3, 6].

2.3 Verified findings versus assumptions

Statement	Status	Basis
Pedestrians frequently experience near-misses at unsignalised crossings	**Verified**	Survey Q3, 91.7 % (n = 36)
Truck/bus blind spots are perceived as a major cause of late detection	**Verified**	Survey Q4, 61.1 % (n = 36)
Users accept automatic camera-based warning at crossings	**Verified**	Survey Q8, 88.6 % strongly support, 0 % oppose
A phone camera + laptop can detect people in a zone with usable precision	**Verified (lab)**	Research-work prototype: precision 0.94, F1 0.92 on 847 lab events; to be re-established on public data (E1)
Foot-point rules beat bounding-box overlap on public datasets	**Assumption**	Hypothesis H1; evaluated in M5 (E1), success criterion in Section 9
Operators prefer no video storage (privacy mode)	**Assumption**	Open survey answers tie acceptance to warning, not surveillance; Law of RK No. 94-V on personal data
The system works outdoors in rain / snow / night	**Assumption**	Night tested in the lab prototype; weather conditions are experiment E7
A real client (yard / depot) will deploy the system	**Not yet**	No agreement with a client; interviews with yard safety staff and drivers are planned for the next period

3. Literature review

The review covers five groups of sources: (1) road-safety statistics and policy, (2) vehicle blind zones and vulnerable road users, (3) passive crossing infrastructure (lighting, LED markings, warning beacons), (4) real-time person detection on edge hardware, (5) geometric and tracking methods used to decide where a detected person is. The full list has 16 sources, of which 12 are peer-reviewed publications; the core detection and edge-computing sources are from 2021–2026, while foundational works (point-in-polygon algorithm, crowd analysis survey) are retained.

3.1 Blind zones of large vehicles are a measurable risk for vulnerable road users

Drake et al. validate a web-based tool that estimates a vehicle's blind zone from the driver's seat and show that large vehicles, heavy trucks in particular, hide pedestrians and cyclists close to the cab; they argue for policy that screens vehicles by blind-zone size [16]. The study measures the vehicle, not the environment: it tells how large the invisible area is, but nothing about whether somebody is standing in it at a given moment. public.safe takes the complementary view — it watches the area on the ground where the blind zone falls (loading bay, nearside of a parked truck, kerb) and reports presence.

3.2 Passive infrastructure improves visibility but does not analyse the scene

Before-after studies of crosswalk warning systems ("SeeMe", Norway) show that pedestrian-activated flashing beacons increase driver yielding [3]; LED-lit crosswalks and integrated lighting-warning systems improve night-time yielding at unsignalised crossings [4, 5]. All three studies measure driver behaviour, not detection: the systems either react to a push button / simple presence sensor or light the crossing permanently. They do not know whether a person is actually in the dangerous area, cannot distinguish a person from a parked bicycle, and cannot be moved to a loading bay or yard. This is the gap public.safe addresses with class-aware detection and a configurable polygon.

3.3 Real-time person detection at the edge is feasible on commodity hardware

YOLO-family detectors dominate real-time surveillance work. Nguyen et al. apply YOLO to real-time human detection on edge devices for smart surveillance [6]; Maltezos et al. report 70 ms per frame for YOLOv5 person detection on embedded GPUs in a distributed edge platform with IoT alert sharing, confirming the latency advantage of edge inference over centralised pipelines [11]. An FPGA deployment of a tiny YOLO reaches real-time detection at 3.5 W [12], and an edge surveillance node with a MobileNet-SSD detector and a VPU accelerator processes up to 12 camera streams at 30 fps [14] — both confirming that detection can live next to the camera on inexpensive hardware. Night-time variants (YOLOv8-night) show that low-light performance is a known weakness that can be mitigated by training data and architecture changes [13]. The US DOT review of cost-effective computer-vision solutions for transportation [15] explicitly recommends consumer cameras as a cheap alternative to stationary sensor networks. Ultralytics YOLOv8 [8] is the detector used here (nano variant: 37.3 mAP on COCO at 3.2 M parameters).

3.4 Deciding where a person is: bounding-box overlap versus ground-contact points

Most zone-intrusion analytics decide membership by intersecting the detection box with the zone (IoU or any-overlap) [6, 7]. The failure mode is systematic: a person standing next to the zone has the upper part of the box over it while the feet are outside, which produces a false alarm. Crowd-analysis work uses ground-plane projection of the box bottom instead [9]; the classical ray-casting point-in-polygon algorithm [10] gives an O(n) exact test for that point. Multi-object trackers such as ByteTrack [7] add temporal identity but still rely on the box geometry for position. The research work that precedes this project found that a three-point foot-contact rule raised precision from 0.71 to 0.94 in laboratory scenarios; however, the published ablation is not consistent with an "any of three points" rule (a superset rule cannot have fewer positives than

its subset), which is why experiment E2 re-establishes the result with all six rules computed on the same frames (ADR 0008).

3.5 What the project takes from the literature

- Use a nano-scale YOLO detector on the laptop GPU/CPU; do not attempt on-phone inference (browser WebGPU inference was excluded as too slow and battery-hungry for continuous use).
- Treat bounding-box overlap as the baseline and foot-anchored rules as the proposed method; report precision/recall with confidence intervals and a paired test (McNemar), following the evaluation style of [11, 14].
- Bound tail latency explicitly (per-camera latest-frame mailbox) rather than relying on average throughput.
- Keep identity out of the pipeline (no faces, no tracking IDs persisted) — privacy requirements in [15] and RK law.

3.6 Structured analysis of sources

#	Source	Type / year	Approach	Key finding	Limitation for this project	Used in
1	WHO, Global Status Report on Road Safety 2023	Report, 2023	Global statistics	Pedestrians = 23 % of road deaths	No site-level data	§1.1, §2.3
2	WHO, Kazakhstan country profile 2023	Report, 2023	National statistics	High road-death rate in KZ	No breakdown by location type	§2.3
3	Hoye & Laureshyn, “SeeMe at the crosswalk”, Transp. Res. F 60, 2019	Journal, 2019	Before-after field study of a pedestrian-activated warning system	Warning beacons increase yielding	No detection; fixed install	§3.2, §4
4	Patella et al., “LED lighting crosswalk”, Safety 6(2), 2020	Journal, 2020	Experimental LED crosswalk	Improves night visibility	Passive only	§3.2
5	Costa et al., “Integrated lighting-warning system”, Transp. Res. F 68, 2020	Journal, 2020	Night-time yielding study	Lighting + warning improves yielding	No scene analysis	§3.2
6	Nguyen et al., “YOLO based real-time human detection for smart video surveillance at the edge”, IEEE ICCE 2021	Conference, 2021	YOLO on edge devices	Real-time human detection feasible	Box-overlap zone logic	§3.3, §3.4
7	Zhang et al., “ByteTrack”, ECCV 2022	Conference, 2022	Multi-object tracking	Associating every box improves tracking	Tracking is P1 (E4)	§3.4, §9
8	Jocher et al., Ultralytics YOLOv8, 2023	Software, 2023	Detector family	YOLOv8n 37.3 mAP, 3.2 M params	AGPL-3.0 licence (ADR 0009)	§5.1
9	Zhan et al., “Crowd analysis: a survey”, MVA 19, 2008	Journal, 2008 (foundational)	Survey of crowd analytics	Ground-plane projection of box bottom	Pre-deep-learning	§3.4

#	Source	Type / year	Approach	Key finding	Limitation for this project	Used in
10	Shimrat, “Algorithm 112: position of point relative to polygon”, CACM 1962	Journal, 1962 (foundational)	Ray-casting PIP	Exact O(n) point-in-polygon test	—	§3.4, geometry.py
11	Maltezos et al., “Video analytics for person detection with edge computing”, Computation 10(3), 2022	Journal, 2022	YOLOv5 on embedded GPU + IoT alerts	70 ms/frame; edge beats central	Dedicated hardware	§3.3, §4
12	Al Amin et al., “FPGA-based real-time object detection and classification using YOLO for edge computing”, IEEE Access 12, 2024	Journal, 2024	Tiny YOLO on an FPGA board	Real-time detection at 3.5 W	Specialised hardware	§3.3
13	Wang et al., “YOLOv8-night”, Electronics Letters 2024	Journal, 2024	Low-light YOLOv8 variant	Night performance can be improved	Different domain (wildlife)	§3.3, risks
14	Cob-Parro et al., “Smart video surveillance system based on edge computing”, Sensors 21(9), 2021	Journal, 2021	MobileNet-SSD + Kalman tracking on an Atom + Myriad X node	12 streams at 30 fps on an edge node; > 80 % detection/counting metrics	Dedicated IP cameras; no zone logic	§3.3, §9
15	US DOT, “Cost-effective computer vision for smart transportation”, 2023	Report, 2023	Review	Consumer cameras are a cheap alternative	US context	§1.5, §3.5
16	Drake et al., “Web-based vehicle blind zone estimation: validation and policy implications”, Frontiers in Future Transportation 4, 2023	Journal, 2023	Field validation of a blind-zone estimation tool	Large vehicles hide VRUs near the cab; blind-zone size is a policy lever	Measures the vehicle, not presence in the zone	§1.1, §3.1

The reference list (Section 12) follows IEEE style; every entry was checked against the publisher or Crossref record in October 2026. Further sources on on-vehicle VRU detection will be added as the driver-warning extension (F-ALERT-1) is designed.

4. Existing system comparison

Five systems that solve a neighbouring problem were compared with the proposed solution along eight criteria. Two groups exist: on-vehicle collision-warning kits (Mobileye Shield+, Brigade Sidescan Predict) and fixed-camera video analytics (Hikvision AcuSense, AXIS Object Analytics, the open-source Frigate NVR). The pedestrian-activated crosswalk warning system “SeeMe” [3] is included as the infrastructure reference. Claims are taken from vendor documentation and the cited studies; where a vendor publishes no figure the cell says so.

Criterion	Mobileye Shield+ (on-vehicle)	Brigade Sidescan Predict (on-vehicle)	Hikvision AcuSense (camera / NVR)	AXIS Object Analytics (edge camera)	Frigate NVR (open source)	SeeMe crosswalk warning [3]	public.safe (proposed)
Sensing	Forward + side cameras on the vehicle, AI hazard analysis	6 ultrasonic sensors, 2.5 m range, nearside	Fixed IP camera, deep-learning human/vehicle filter	Fixed Axis camera with on-camera DLPU	Any RTSP IP camera + server (CPU / Coral / GPU)	Presence sensor / push button at the crossing	Any smartphone browser camera, no install
What is detected	Pedestrians, cyclists, vehicles ahead and in side blind spots	Objects in the blind spot (moving / static); no class	Humans, vehicles	Humans, vehicles, vehicle types	COCO classes (person, car, ...)	Presence only	Persons (YOLOv8n class 0)
Zone / geometry logic	Trajectory and time-to-collision, vehicle-centred	Predicted collision from speed, direction, turning rate	Intrusion box, line crossing, region enter/exit — box-based, up to 4 regions	Object in area, line crossing, time in area, occupancy	Polygon zones; membership = bottom-centre of the box, with inertia frames	None	Polygon zones; 6 rules every frame (foot-point variants + bbox overlap), hysteresis state machine
Protects	One equipped vehicle	One equipped vehicle (rigid body, off above 30 km/h)	One fixed site per camera	One fixed site per camera	Fixed cameras at one site	One crossing	Any temporary or fixed site; several phones per laptop
Deployment unit and cost	Retrofit kit per vehicle, professional install; price on quote (vendor cites ROI < 12 months)	Retrofit kit, ECU, display, cabling; price on quote	Camera + NVR, cabling, licences; site-level CCTV budget	Axis cameras only (MLPU/DLPU), analytics free with the camera	Free software; needs a server and a Coral/GPU for several cameras	Pole, beacons, sensor, power; civil works	Existing phones + one laptop; software free (AGPL-3.0)

Criterion	Mobileye Shield+ (on-vehicle)	Brigade Sidescan Predict (on-vehicle)	Hikvision AcuSense (camera / NVR)	AXIS Object Analytics (edge camera)	Frigate NVR (open source)	SeeMe crosswalk warning [3]	public.safe (proposed)
Time to deploy	Days (installation per vehicle)	Days	Days–weeks (cabling, mounting)	Days	Hours (if cameras exist)	Weeks (construction)	Minutes: scan a QR, draw a polygon
Openness / reproducibility	Closed; no accuracy data published	Closed; no detection-rate data published	Closed; no accuracy data published	Closed; no accuracy figures published	Open source (MIT); zone rule documented	Peer-reviewed field study	Open source; same code runs offline on MOT17 with pre-registered metrics
Privacy	In-vehicle recording	No video	Video recorded on NVR	Video recorded / streamed	Records clips and snapshots	No video	No video stored; privacy mode sends JSON only
Published latency	Not published	Not published	Not published	Not published	Not published (depends on hardware)	n/a	Measured: 145 ms median, 380 ms p99 (lab, 3 phones); re-measured in E5/E6

4.1 Conclusions from the comparison

- On-vehicle kits protect only the vehicle they are installed in and cost a retrofit per vehicle; they cannot cover a yard where many third-party trucks operate. A **site-centred** system that watches the dangerous area itself is complementary and much cheaper.
- Fixed-camera analytics decide membership by box intersection (Hikvision, Axis) or by the box bottom-centre (Frigate). None of them lets the operator choose and compare rules, and none publishes precision/recall — which is why the project treats the rule itself as the research object and measures it.
- Only Frigate is open and reproducible, but it needs installed IP cameras and a server; it has no pairing flow for ad-hoc phone cameras and no offline evaluation on public datasets.
- Crosswalk warning systems prove that a visible signal changes driver behaviour [3, 5]; public.safe produces the event that such a signal needs, from a camera instead of a push button.

5. Technology comparison

Seven key decisions were compared against 2–3 alternatives each. Decisions marked ADR are recorded in `docs/adr/` with context, alternatives and consequences.

5.1 Object detector

Model	Year / framework	COCO mAP50-95	Params / FLOPs	CPU ONNX latency (640 px)	Licence	Assessment for public.safe
YOLOv3	2018, Darknet / PyTorch ports	≈ 33 (608 px)	62 M / 140 GFLOPs	Hundreds of ms on CPU	Various ports	Too heavy for a laptop serving several phones; outdated tooling
YOLOv7	2022, PyTorch	51.4 (full); tiny ≈ 37–38	37 M / 105 GFLOPs (full)	Full model not CPU-real-time; tiny comparable to v8n	GPL-3.0	Strong accuracy, but research-style repo, no maintained pip package, GPL
YOLOv8n (chosen)	2023, Ultralytics PyTorch	37.3	3.2 M / 8.7 GFLOPs	80 ms (vendor); 49 ms FP32 on Apple MPS measured in M4	AGPL-3.0	Maintained package, MPS/CUDA/CPU auto-selection, known behaviour from the research work; licence handled by ADR 0009
YOLO11n	2024, Ultralytics	39.5	2.6 M / 6.5 GFLOPs	56 ms (vendor)	AGPL-3.0	Better and lighter; drop-in replacement behind the `Detector` seam — candidate for E7 re-run after the baseline is established with v8n

YOLOv8n was kept because the research hypotheses were formulated and partly measured with it, so results stay comparable; the detector seam (ADR 0009) makes swapping to YOLO11n or to a permissively licensed ONNX model a one-file change. FP16 on MPS was measured and found *slower* (57.6 vs 49.2 ms per frame, identical boxes), so FP32 is the default (`docs/EVALUATION.md`).

5.2 Server platform

Option	Pros	Cons	Decision
FastAPI + Uvicorn (Python 3.12)	Native async WebSockets, Pydantic validation, same language as the CV stack (PyTorch, OpenCV, Shapely), OpenAPI for free	Single process; GIL means CV runs in a worker thread	**Chosen** — CV code and API in one process, no serialisation between them
Flask / Django	Mature, large ecosystem	WebSockets need extra layers (Channels, Socket.IO); sync by default	Rejected
Node.js (Express / NestJS)	Excellent WebSocket support	Inference would need a second Python process or ONNX runtime in JS; two codebases for core logic	Rejected

5.3 Phone → server transport

Option	Pros	Cons	Decision
Binary WebSocket (WSS), JPEG per frame	Works in every mobile browser, simple backpressure (one unsent frame), frames are already images for the detector, first-message auth (ADR 0004)	No inter-frame compression → ~30–80 KB per frame at quality 65	**Chosen**
WebRTC video track	H.264 compression, adaptive bitrate	Needs a media server or aiortc, decoding on the server, signalling, TURN; much more complexity for a LAN	Deferred (behind the same `FrameSource` interface)
HTTP POST / MJPEG per frame	Trivial	Connection overhead per frame, no server→phone control channel	Rejected

5.4 Data storage (ADR 0003)

Option	Pros	Cons	Decision
SQLite (WAL) by default	Zero setup, one file, fast enough for a few rows per second, migrations via Alembic	Single-writer; not for multi-process deployments	**Chosen default**
PostgreSQL via `DATABASE_URL`	Multi-process, JSONB, native UUID	Needs a service (Docker Compose provided)	**Supported option**; migration round-trip tested on both
Redis / message broker + DB	Decoupled workers	Extra infrastructure, excluded by non-goals	Rejected

5.5 Frame scheduling under several phones (ADR 0006)

Option	Pros	Cons	Decision
Per-camera latest-frame mailbox + round-robin worker	Each camera keeps only its newest frame, bounded memory, fair share per camera, latency bounded by one round	Frames are dropped on purpose (counted as `dropped_replaced`)	**Chosen**
Shared FIFO queue	Simplest	Queue grows under load, stale frames, a fast phone starves the others (lab: p99 643 ms vs 380 ms)	Kept only as experiment baseline (E5)
One worker thread per camera	Isolation	GPU/MPS is shared anyway; contention and memory grow with cameras	Rejected

5.6 Device authentication (ADR 0005) and rate limiting (ADR 0002)

Option	Pros	Cons	Decision
Opaque 256-bit random device tokens, stored as SHA-256	Instant revocation, no signing key, one indexed lookup, never in URLs/logs	Server-side state (one row per camera)	**Chosen**
Signed JWT	Stateless	Revocation needs a deny-list anyway; leaked token valid until expiry; key management	Rejected
In-process sliding-window limiter vs slowapi	Deterministic tests with injected clock; no extra dependency	Per-process (acceptable on one host)	**Chosen** (ADR 0002)

5.7 Frontend

Option	Pros	Cons	Decision
React 19 + TypeScript + Vite, TanStack Query	Typed API client, canvas overlays easy, Playwright tooling, largest ecosystem	Bundle size larger than Svelte	**Chosen** — same pages serve the dashboard and the phone (pair/camera)
Vue 3	Simple reactivity	No advantage for canvas-heavy UI	Rejected
Svelte	Smallest bundle	Smaller ecosystem for testing/tooling	Rejected
Server-rendered overlays (JPEG with boxes)	Dumb client	One encode per subscriber, no privacy mode, pixels instead of data	Rejected (ADR 0007: overlays drawn client-side from JSON)

6. Requirements specification

Requirement IDs are the ones used in the implementation specification (`docs/IMPLEMENTATION_SPEC.md`, Sections C–D), in test names and in milestone acceptance. Priority follows MoSCoW: **M** must, **S** should, **C** could. "Verified by" names the automated suite or manual check that proves the criterion.

6.1 Functional requirements

ID	Requirement	Pr.	Acceptance criteria	Verified by	Status
F-ADM-1/2	Single admin account; Argon2id password hash from `.env`; server-side session cookie (HttpOnly, Secure, SameSite=Strict, 12 h idle)	M	Server refuses to start without a hash and prints the CLI command; login/logout work; password rotation invalidates all sessions	unit/test_auth, integration/test_api_auth	Done (M1)
F-SES-1/2	Create, list, rename, archive monitoring sessions; session has 0..N cameras, camera has 0..N zones	M	CRUD endpoints return the documented shapes; archived sessions reject new phones	integration/test_api_sessions	Done (M1)
F-PAIR-1/2	Admin mints a pairing PIN (5 min lifetime); phone redeems it once and receives a camera id and device token	M	PIN single-use (two concurrent redeems → one wins); lockout after max attempts; PIN never appears in any response except mint	unit/test_pairing, security/test_pairing_security	Done (M2)
F-CAM-1/2/3	Camera statuses paired/online/offline/paused/error/revoked; settings (fps 1–30, JPEG 30–95, long edge 640–1920, confidence, privacy mode) pushed to the phone within 2 s	M	Offline detected within 12 s; settings message arrives on the camera socket after PATCH; list shows fps, drop ratio, RTT, occupancy	integration/test_api_cameras, test_ws_camera	Done (M2–M3)

ID	Requirement	Pr.	Acceptance criteria	Verified by	Status
F-PHONE-1…7	Phone page: secure-context check, getUserMedia rear camera, capture at target fps, ≤ 1 unsent frame, reconnect with backoff, pause on hidden tab, ping/pong RTT, “forget this phone”	M	Page never shows a PIN form that cannot succeed; streaming resumes after a reload; close code 4003 stops and clears the token	e2e/pairing.spec.ts, manual iOS checklist	Done (M3); Android unverified
F-ZONE-1…4	Polygon editor on the live frame (3–20 vertices), stored in normalised coordinates, validated as a simple polygon; settings active_rule, enter_frames, exit_frames, enabled; needs_review on aspect change	M	Invalid polygons rejected with 422; saving and reloading reproduces the same pixels; editor usable with mouse and keyboard	unit/test_geometry, integration/test_api_zones, e2e/zones.spec.ts	Done (M4)
F-CV-1	Person detection with YOLOv8n, class 0 only, device auto-selected (mps → cuda → cpu)	M	Golden image yields ≥ 1 box with conf ≥ 0.3 inside the fixture zone; model SHA-256 shown on Health	cv/test_model_golden (make test-model)	Done (M4)
F-CV-2	All six membership rules computed for every detection and zone, every frame; active rule drives the state machine	M	Invariants `all ⊆ majority ⊆ any ⊆ bbox_overlap` and `bottom_center ⊆ any` hold on 1 000 random cases; per-rule counts in every frame result	unit/test_rules (Hypothesis)	Done (M4)
F-CV-3/4	Zone state machine with hysteresis; events zone_entered / zone_cleared / occupancy_changed persisted and pushed within 100 ms	M	1-frame blip → no event; 3 frames → entered; flicker stays occupied until exit_frames; exactly one entered and one cleared per visit	unit/test_state_machine, cv/test_pipeline_events	Done (M4)

ID	Requirement	Pr.	Acceptance criteria	Verified by	Status
F-CV-5	Dashboard overlays: boxes with confidence, three bottom anchors, zone polygon green/red, per-zone count — drawn client-side from JSON	M	Overlay scales correctly when the server down-scaled the frame; privacy mode shows overlays without video	e2e screenshots, manual	Done (M4)
F-DASH-1…4	Pages: login, sessions, session detail, camera detail, events, security, health; live view via dashboard socket; loading/empty/error states; metrics panel	M	Every list has loading, empty and error states; preview rate ≤ processed fps; metrics every 2 s	e2e suite, Playwright screenshots	Done (M2–M4)
F-EXP-1	CSV export of filtered events and per-camera metric samples, one column per rule	S	CSV columns match Section K of the spec; export respects filters	integration/test_api_events	Done (M4)
F-EVAL-1/2	Evaluation CLI runs detector + rules + state machine on a video or image sequence with zone and ground-truth files; recorder writes frames and results for re-evaluation	M	Outputs results.csv, per_frame.jsonl, metrics.json, report.md; metrics on a synthetic 100-frame sequence equal hand-computed values	cv tests (planned), eval CLI	In progress (M5, #13)
F-EVAL-3	Load test against a live server with 1–5 fake phones, reporting fps, drop ratio, span percentiles and fairness ratio	S	Fairness ratio ≤ 1.3 under “latest”; memory growth < 20 %	tests/load/run_loadtest.py	Done (M3)
F-SEC-1/2/3	Admin routes and dashboard socket require the session; camera socket requires a device token; rate limits 5/min login, 10/min + 30/10 min redeem; security log	M	429 with Retry-After; login timing for unknown vs wrong password within 20 %; tokens never in logs or responses	security/* (8 tests)	Done (M1–M2)

ID	Requirement	Pr.	Acceptance criteria	Verified by	Status
F-PRIV-1/2/3	No frame written to disk unless recording is enabled; privacy mode per camera; persistent “this phone is streaming” notice	M	grep of data/ after a session shows no images; privacy-mode camera sends no binary to dashboards	integration/test_ws_dashboard	Done (M4)
F-ALERT-1	External alert output (webhook / GPIO) on zone_entered for a driver-facing light	C	Configurable endpoint called within 200 ms of the event	—	Planned (after M6)

6.2 Non-functional requirements

ID	Area	Requirement and acceptance criterion	Pr.	Verified by
NF-SEC	Security	HTTPS/WSS only (mkcert locally); Argon2id; SHA-256 for tokens/PINs/session ids with salted PINs; constant-time comparison; CSRF via SameSite + X-Requested-With; CSP when FastAPI serves the SPA; no secrets in logs or repo	M	security suite; CSP header test; .env git-ignored
NF-LAT	Latency	Server-side receive → dashboard-send p50 ≤ 150 ms and p99 ≤ 400 ms with 3 phones at 10 fps on an Apple Silicon laptop. Goal for the metrics panel; acceptance = numbers are measured and reported correctly	S	E5/E6 load test and metrics export
NF-THR	Throughput	3 phones × 10 fps processed without unbounded memory; 5 phones degrade gracefully (drops counted, no crash)	M	make test-load
NF-REL	Reliability	Phone disconnects, server restarts and worker exceptions never lose persisted data; a worker exception marks one camera error and others continue; 8 h run with < 20 % memory growth	M	integration tests; soak test (M6)
NF-PERF	Performance	Decode + inference + rules + fan-out for a 1280×720 frame ≤ 60 ms p50 on MPS (reported, not a gate); CPU fallback works at reduced fps	S	timings in frame_result; make test-model
NF-MAINT	Maintainability	mypy strict on `publicsafe/core`; ruff, black, eslint, prettier, tsc clean; core importable without FastAPI; ADR for every listed decision; core coverage ≥ 90 %, overall ≥ 75 %	M	make lint; coverage report
NF-PORT	Portability	macOS Apple Silicon primary; Linux x86 with CPU or CUDA runs the test suite (CI-ready); Windows not required	S	make test on Linux runner
NF-PRIV	Privacy	Logs contain camera ids and sizes, never frame contents; events and metrics retained until the admin deletes the session	M	log grep test
NF-ACC	Accessibility	Dashboard keyboard-navigable; colour is never the only status indicator	S	manual check, Playwright
NF-OBS	Observability	Structured JSON logs with request and camera ids; /api/health and /api/metrics	S	M6

7. Architecture and diagrams

public.safe is a three-tier system on one local network: phones capture, one laptop processes, browsers observe. The diagrams below are drawn in PlantUML (C4 notation for context and component); sources are in `docs/diagrams/*.puml` and exports in `docs/diagrams/export/` so they can be edited with any PlantUML tool. Architecture decisions are recorded in nine ADRs (`docs/adr/`), listed in Section 7.6.

7.1 System context

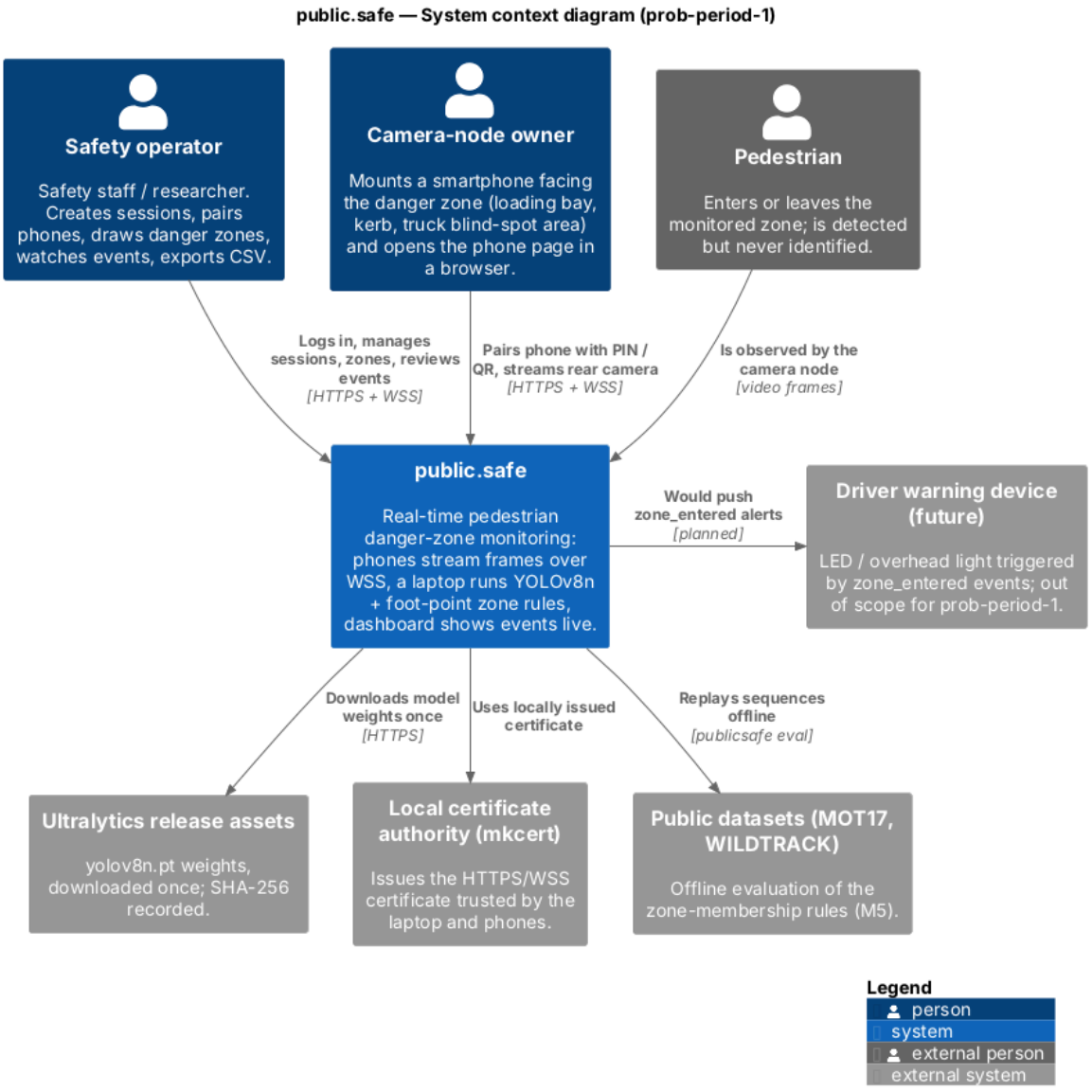
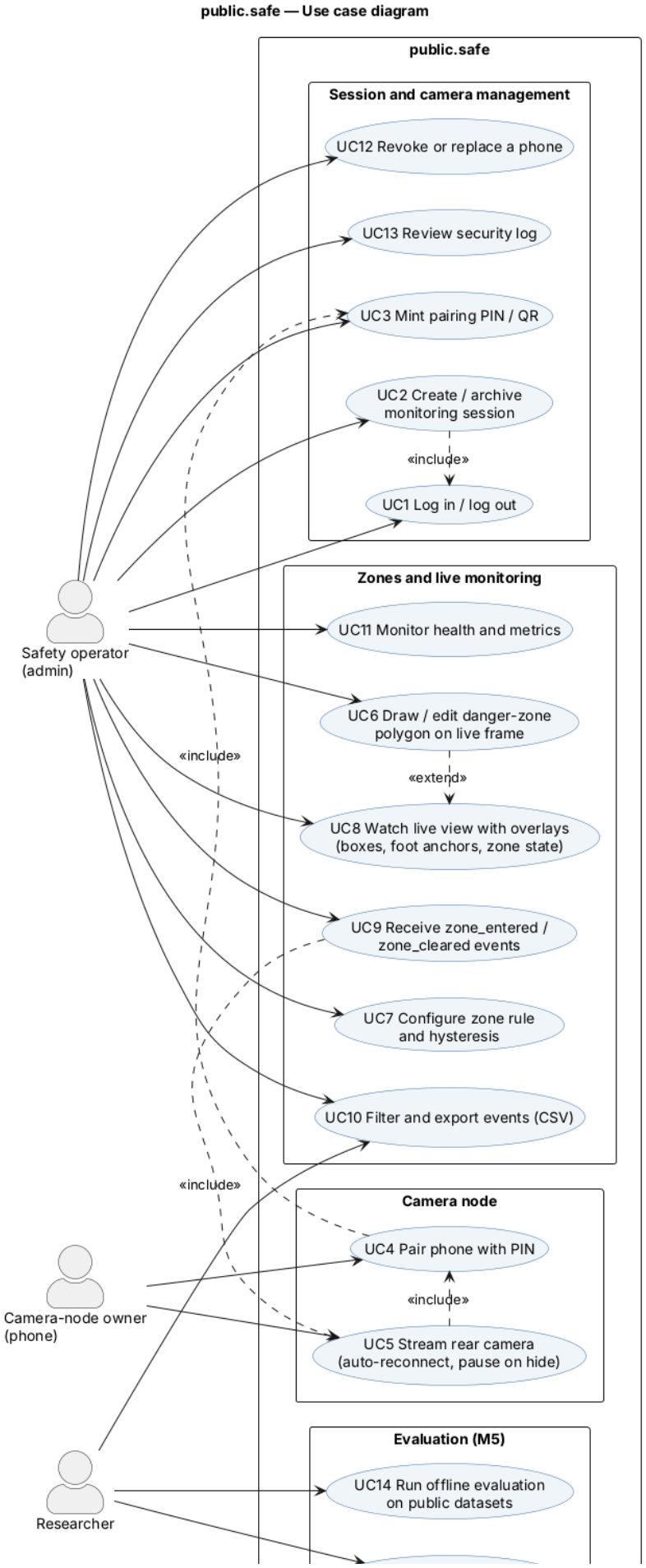


Figure 7.1. System context (C4 level 1). The driver-warning device is a planned external system.

7.2 Use cases


public.safe — Use case diagram
public.safe
Session and camera management
UC12 Revoke or replace a phone
UC13 Review security log
UC3 Mint pairing PIN / QR
UC2 Create / archive monitoring session
«include»
UC1 Log in / log out
Zones and live monitoring
UC11 Monitor health and metrics
UC6 Draw / edit danger-zone polygon on live frame
«extend»
UC8 Watch live view with overlays (boxes, foot anchors, zone state)
UC9 Receive zone_entered / zone_cleared events
UC7 Configure zone rule and hysteresis
UC10 Filter and export events (CSV)
Camera node
UC4 Pair phone with PIN
«include»
UC5 Stream rear camera (auto-reconnect, pause on hide)
Evaluation (M5)
UC14 Run offline evaluation on public datasets
Safety operator (admin)
Camera-node owner (phone)
Researcher
«include»
«include»

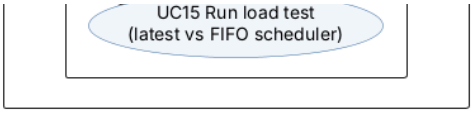


Figure 7.2. Use case diagram: three actors, fifteen use cases grouped by area.

7.3 Component diagram (draft)

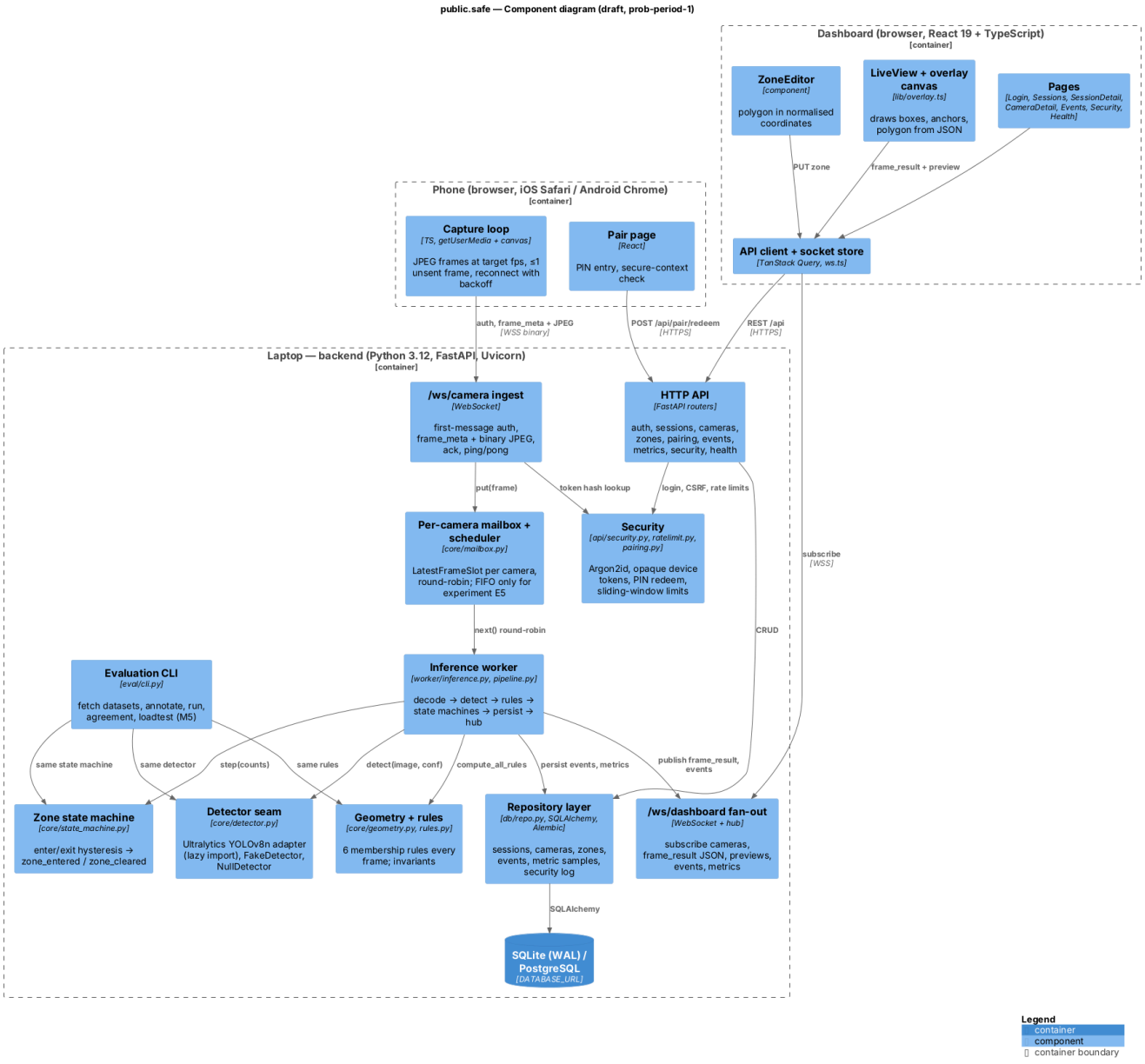


Figure 7.3. Components in the three containers (phone, laptop backend, dashboard). The evaluation CLI reuses the detector, rules and state machine unchanged, which is what makes offline results comparable with live ones.

Processing path. The phone encodes a JPEG and sends it over the camera WebSocket. The ingest handler puts it into that camera's *LatestFrameSlot* without decoding; an unread older frame is replaced and counted. One inference worker thread takes frames round-robin across cameras, decodes, runs YOLOv8n (person class only), computes all six membership rules for every detection and every enabled zone, feeds the active rule's count into the zone state machine, persists events and metric samples, and publishes a `frame_result` JSON (plus the JPEG unless privacy mode) to subscribed dashboards. The dashboard draws overlays on a canvas from that JSON. Every stage is timed (`timings` in the frame result) so latency can be attributed.

7.4 Data model

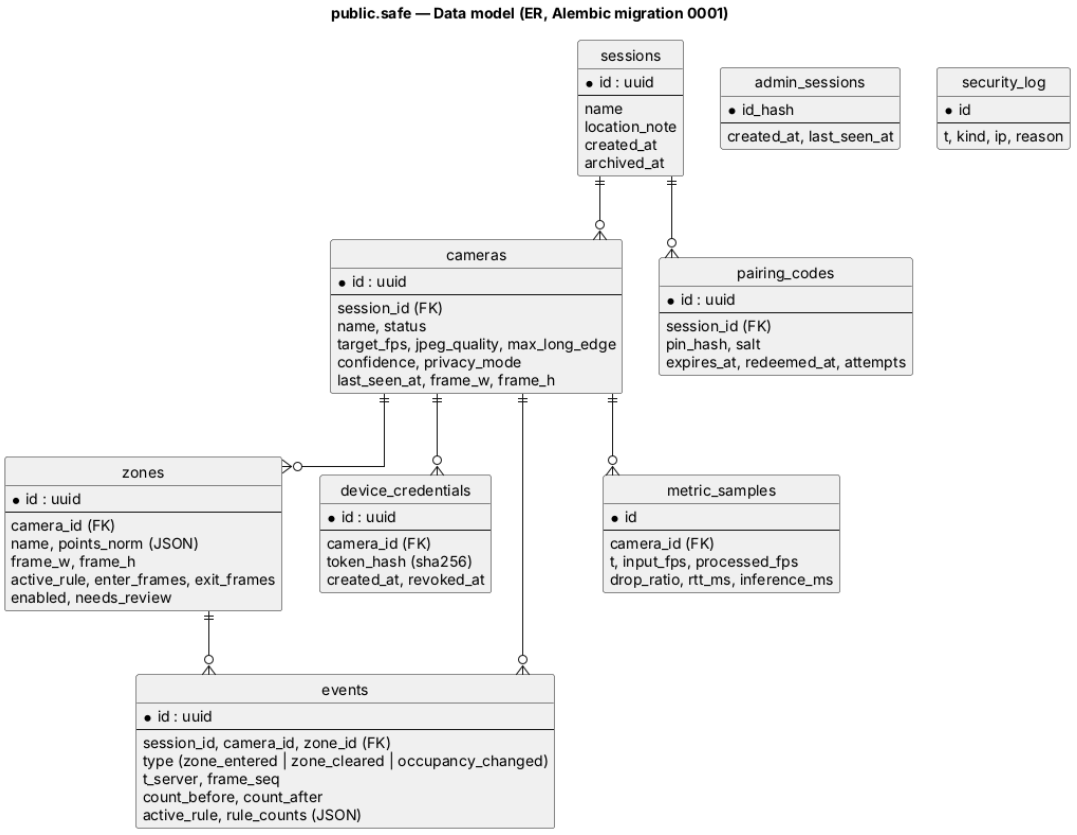


Figure 7.4. Entity–relationship diagram (Alembic migration 0001). Tokens and PINs are stored only as hashes; event rows never contain images.

7.5 Pairing, streaming and event flow

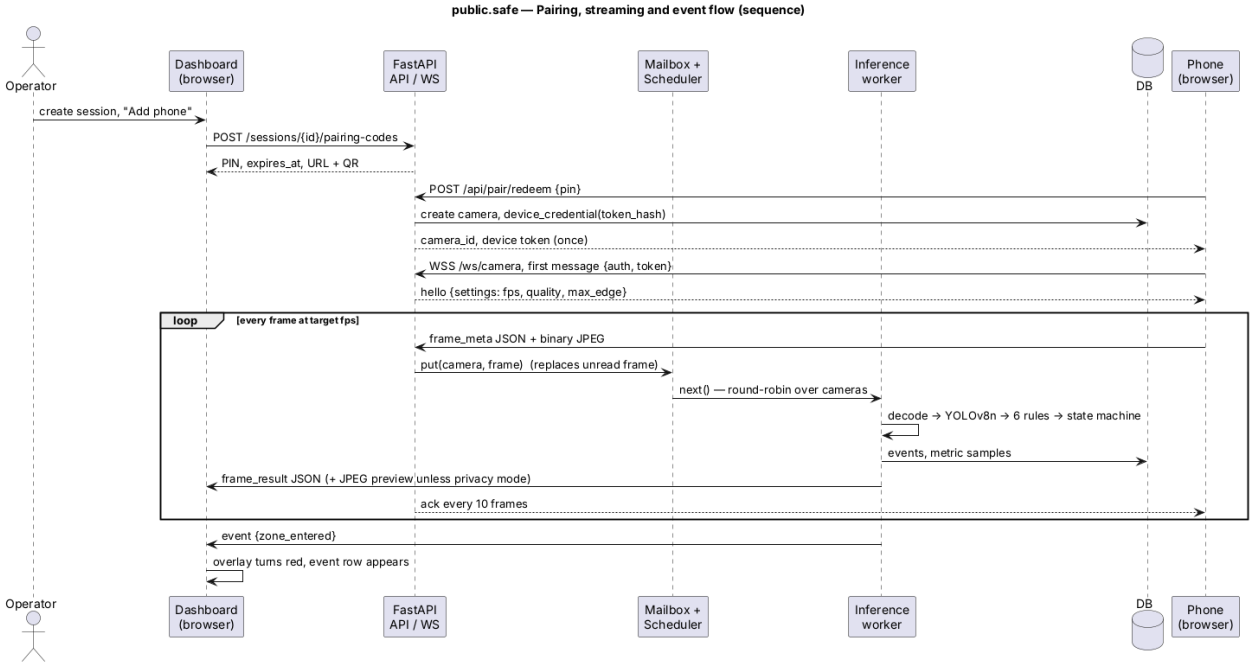


Figure 7.5. Sequence from minting a PIN to a `zone_entered` event on the dashboard.

7.6 Architecture decision records

ADR	Decision	Milestone
0001	Python 3.12 via uv, fully pinned freeze, headless OpenCV only	M0
0002	In-process sliding-window rate limiter instead of slowapi	M1
0003	SQLite by default, PostgreSQL by DATABASE_URL	M1
0004	WebSocket authentication by first message, after accept	M2
0005	Opaque random device tokens instead of JWT	M2
0006	Per-camera latest-frame mailbox with a round-robin scheduler	M3
0007	Overlays are rendered client-side from JSON	M4
0008	All membership rules computed every frame	M4
0009	AGPL-3.0 Ultralytics behind the Detector seam; repository licensed AGPL-3.0	M4

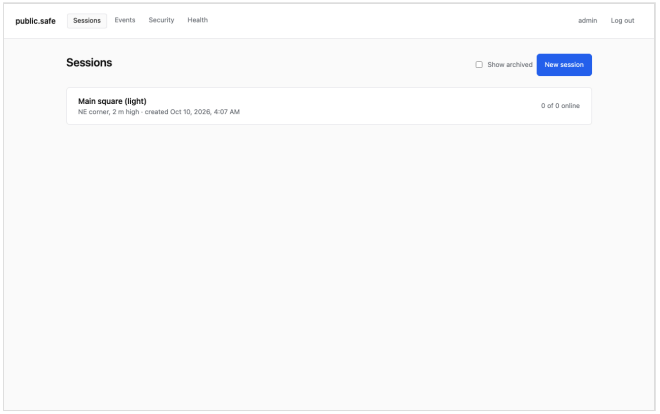
Alignment check: every functional requirement in Section 6 maps to a component in Figure 7.3 (pairing → Security + HTTP API; capture → Capture loop + ingest; zones → ZoneEditor + Geometry; events → State machine + Repository + fan-out), every entity in Figure 7.4 is written by exactly one component, and every use case in Figure 7.2 is exercised by at least one automated test (Section 9.1).

8. UX / prototype

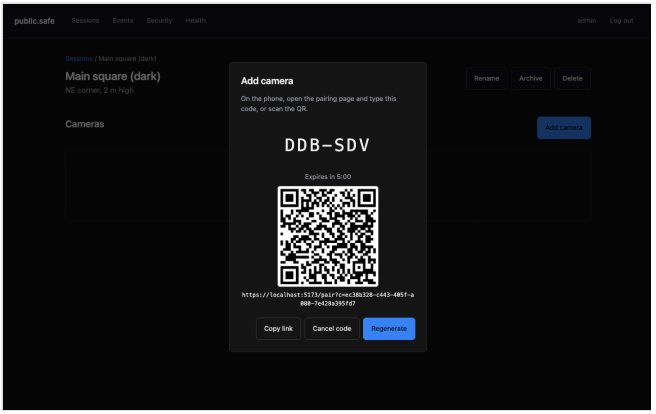
The prototype is a running system, not a mock-up: milestones M0–M4 are implemented and tagged. Screenshots below were taken by the Playwright suite in Chromium against the test backend (the green picture is Chromium's synthetic camera standing in for a phone) and from the earlier research prototype on a real iPhone at night. The repository contains run instructions (`README.md`, `docs/SETUP_MACOS.md`, `docs/PHONE_SETUP.md`).

Prototype access. Repository: github.com/BekdauletBol/public_final/tree/prob-period-1 · Screenshots: `docs/screenshots/m2…m4/` · Run: `make setup && make certs && make dev` → `https://localhost:5173`. The system runs on a local network only; a live demo is shown at the pre-defence. **[optional: link to a demo video]**

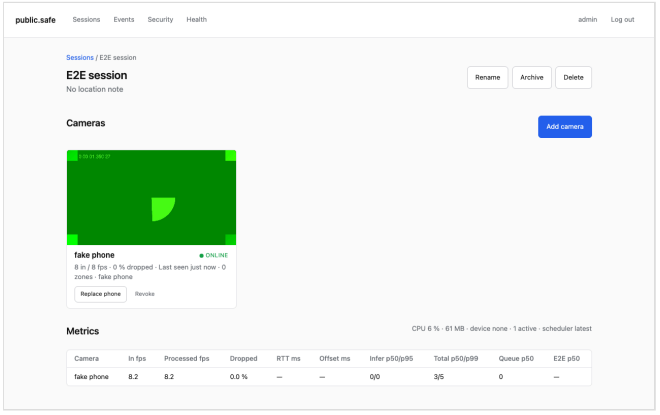
8.1 Key user scenarios



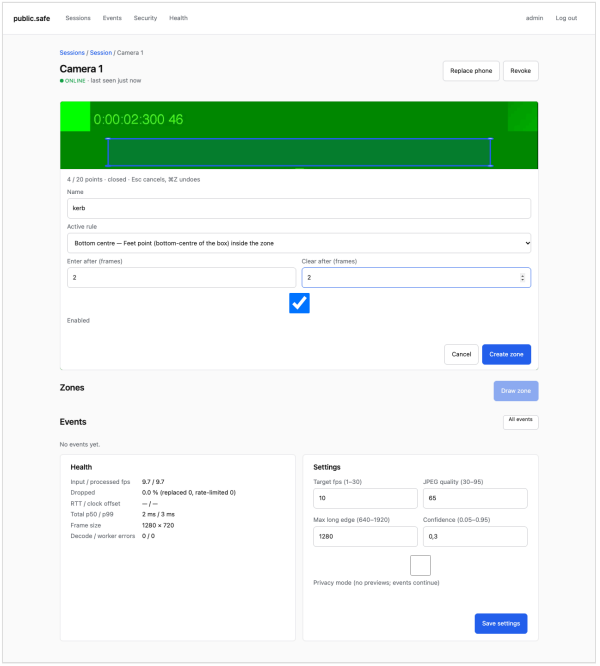
Sessions list (light theme).



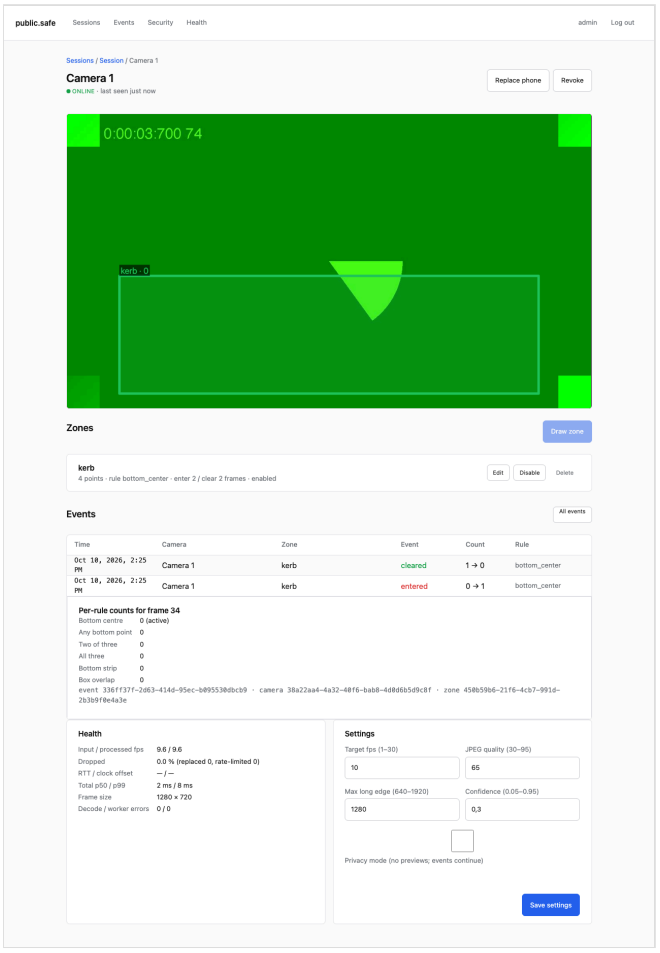
Pairing modal: PIN, link and QR (dark theme).



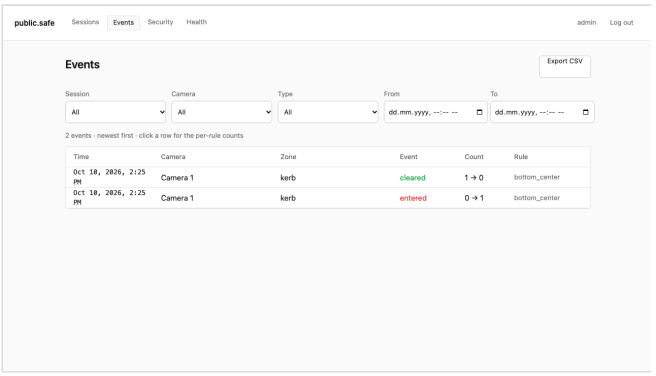
Session page: live camera tile, status, per-camera metrics.



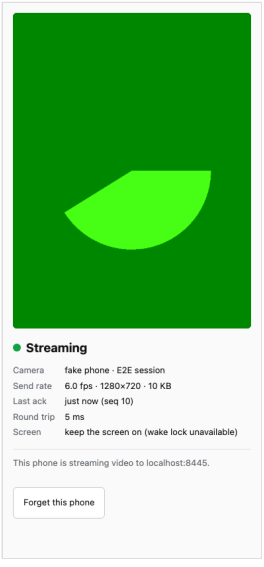
Zone editor: polygon on the live frame, rule and hysteresis settings.



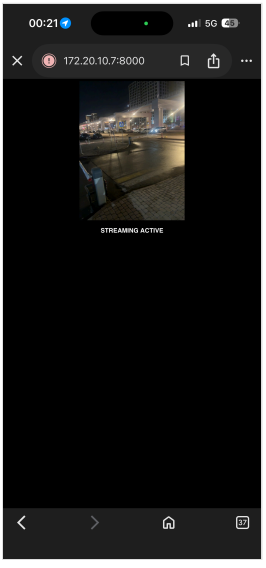
Camera page: zone overlay, one entered and one cleared event, per-rule counts.



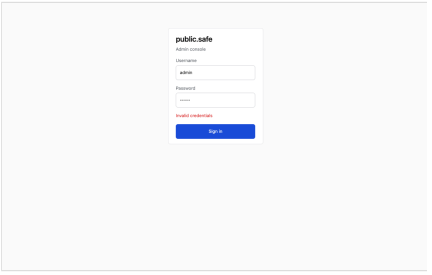
Events page: filters by session, camera, type, time; CSV export.



Phone page while streaming (current version).

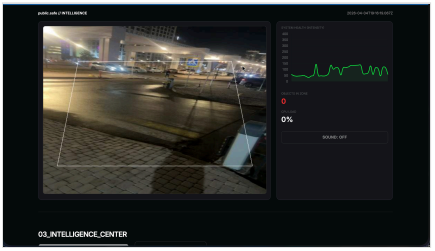


Research prototype on iPhone (previous version).

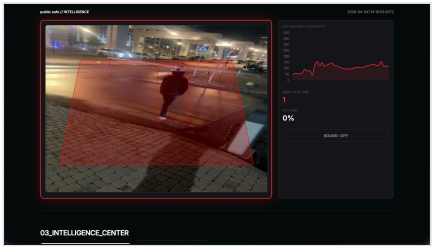


Login with error state.

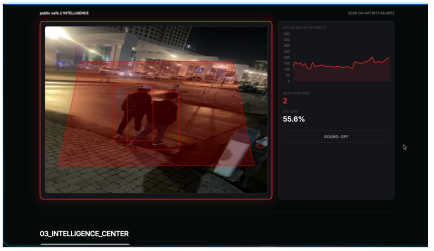
8.2 Detection in real conditions (research prototype, night)



Empty scene: no detections, zone stays safe.



One person inside the zone: alert state.



Two persons counted inside the zone.

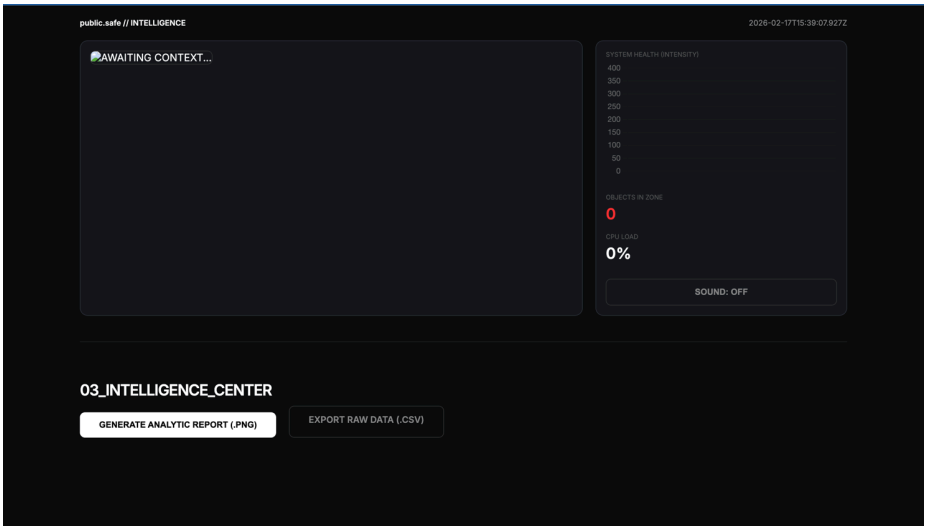


Figure 8.1. Previous dashboard version (research prototype): analytics panel and violation alert. The current version replaced it with the React dashboard above.

8.3 API and data examples

Besides the UI, external systems interact through the REST API and the dashboard WebSocket. Example of a frame result delivered to dashboards (and written by the evaluation CLI line-by-line):

```
{"type":"frame_result","camera_id":"c1…","seq":1842,"w":1280,"h":720,
 "detections":[{"bbox":[412,180,560,640],"conf":0.83,
                "anchors":{"bl":[412,640],"bc":[486,640],"br":[560,640]}}],
 "zones":[{"zone_id":"z1…","active_rule":"bottom_center","state":"occupied",
           "debounced_count":1,
           "counts":{"bbox_overlap":1,"bottom_center":1,"three_point_any":1,
                     "three_point_majority":1,"three_point_all":0,"bottom_strip":1}}],
 "timings":{"decode_ms":4.1,"infer_ms":49.2,"rules_ms":0.3,"total_ms":56.9}}
```

Event row (CSV export): `t_server, session, camera, zone, type, count_before, count_after, active_rule, bbox_overlap, bottom_center, three_point_any, three_point_majority, three_point_all, bottom_strip`. Evaluation command: `publicsafe eval run --config configs/e1_mot17.yaml`.

8.4 Concept of a field installation (future)



Figure 8.2. Concept visualisation (AI-generated illustration, not a prototype): camera node, ground-level LED indicator and overhead driver alert connected to the dashboard. Hardware is out of scope for the diploma; the event interface that would drive it is F-ALERT-1.

9. Verification and evaluation plan

9.1 Testing plan (software verification)

218 automated tests exist at the tagged version; all run locally without network or GPU except the two real-model tests. Every milestone was accepted only when its suite passed (`make test`, `make lint`).

Level	What is checked	Tooling	Tests
Unit	Geometry anchors and all six rules on hand-built cases (box straddling the top edge → only bbox_overlap positive), property test of rule invariants on 1 000 random boxes/polygons, cv2.pointPolygonTest cross-check; state machine sequences; mailbox fairness and drop counting; pairing (single use, lockout, concurrency); rate limiter with injected clock; config	pytest, Hypothesis	136
Integration	Every API endpoint happy path and validation errors, cookie auth, CSRF header, CSV shape; camera socket close codes (4001/4003/4004/1009/4000), ack cadence, ping/pong offset; dashboard subscribe, frame_result, event, privacy mode; Alembic round-trip on SQLite and PostgreSQL; scripted fake phone	pytest-asyncio, httpx, Starlette TestClient	63
Security	429 with Retry-After; login timing unknown-vs-wrong within 20 % over 50 runs; tokens and PINs never in logs, responses or rows; CSP and cookie flags	pytest	8
Computer vision	Scripted 60-frame sequence → expected events; zone payloads; recorder; golden image with real YOLOv8n (≥ 1 box inside the fixture zone); FP16 vs FP32 timing	pytest, make test-model	8
End-to-end	Login → session → PIN → pair in a second browser context with a fake camera → streaming → tile online → draw polygon → event appears	Playwright, Chromium	3
Load	3 fake phones at 15 fps against a live server with a 20 ms fake detector: fairness ratio ≤ 1.3, memory growth < 20 %, p99 queue wait reported; repeated with FIFO to show growth	Python, psutil	1

Manual checks: iOS Safari pairing and streaming checklist (`docs/PHONE_SETUP.md`); Android Chrome is supported by design but not yet verified on a device (risk R3).

9.2 Outcome metrics

- **Per-person zone membership** (each matched detection × rule): TP/FP/FN/TN → precision, recall, F1, FPR; detector misses reported separately as `fn_missed_detection` so rule errors are distinguishable from detector errors.
- **Event level**: zone_entered events matched within ±10 frames → event precision/recall/F1, timing error median and p90, duplicate-event rate.
- **Statistics**: 5 000-sample bootstrap 95 % CIs (seed 1234); exact McNemar test on paired per-person outcomes for each pair of rules; boundary analysis — error rate per 10 px bin of the ground-truth foot distance to the polygon edge.
- **Latency**: server-side spans (queue wait, decode, inference, rules, fan-out) as p50/p90/p95/p99; end-to-end estimate from phone timestamps with RTT-based uncertainty; fairness ratio = max ÷ min processed fps across cameras; drop ratio.

9.3 Baselines, datasets and measurement conditions

Baseline rule	`bbox_overlap` — any intersection of the person box with the zone, the rule used by commercial analytics [6, 7]. Secondary baseline: `bottom_center` (one foot point, as in Frigate).
Baseline scheduler	Shared unbounded FIFO queue (`SCHEDULER_STRATEGY=fifo`), the design of the research prototype.

Datasets	**MOT17** train sequences 02, 04, 09 (1920×1080, 25–30 fps, full-body ground-truth boxes; 04 is a night street from an elevated viewpoint) — primary; **WILDTRACK** (7 calibrated cameras, ground-plane positions) — geometric check with a metric zone; **own recordings** via `EVAL_RECORD=1` for the four lab scenarios plus night, occlusion and tangential approach, with consent of people filmed.
Zone ground truth	Three polygons per sequence (strip, corner, irregular) committed as JSON; per-person `in_zone` labels produced with `publicsafe eval annotate` every 5th frame by two annotators; Cohen's κ reported; auto-derived labels used only for sanity checks.
Conditions	Detector YOLOv8n, conf 0.30, imgsz 640, FP32; weights SHA-256 recorded; seeds fixed; `run.json` with host, device, package versions, git commit and config hash in every output directory. Live tests: Apple Silicon laptop, 5 GHz Wi-Fi, phones at 10 fps, 1280×720, JPEG quality 65.

9.4 Pre-registered experiments and success criteria

Id	Question	Success criterion (fixed before running; copied from `docs/EVALUATION.md`)	Status
E1	Foot-anchored rules vs bbox overlap	At least one of bottom_center / three_point_* has precision ≥ bbox_overlap + 0.10 with non-overlapping 95 % CIs and recall loss ≤ 0.05, on ≥ 2 of 3 data sources	Harness in progress (#13, #14)
E2	Which 3-point combination; invariants	All rule results reported; invariants hold on every sequence; active rule recommended by F1 with boundary analysis	Planned
E3	Hysteresis effect on duplicate events	Duplicate-event rate falls ≥ 50 % with 3-in/5-out vs none; timing error p90 ≤ 10 frames	Planned
E4	Tracking (optional)	Event F1 improves or is unchanged; reported either way	Optional
E5	Latest-frame mailbox vs FIFO under load	With 3 cameras p99 queue wait under latest ≥ 30 % lower than FIFO; fairness ratio ≤ 1.3	Load test exists; benchmark #15
E6	Live latency with real phones	p50 reported; e2e estimate with RTT uncertainty; ≤ 200 ms p50 is a goal, not a gate	Needs 2–3 phones
E7	Difficult conditions (night, rain, occlusion)	Per-condition recall and FPR tabulated; descriptive	Planned

How the project aim will be demonstrated: (1) the live system at the pre-defence with a phone, a zone and events; (2) E1/E2 tables with confidence intervals showing whether the foot-point rule reduces false alarms on public data; (3) E5 showing bounded tail latency; (4) the test suite and lint passing on the tagged commit.

10. Project management and risks

10.1 Repository, task tracking and workflow

Git repository	github.com/BekdauletBol/public_final — public, AGPL-3.0, 28 commits from 10 Oct to 18 Oct 2026, one branch (`main`) with milestone-scoped commits, tag `prob-period-1`
Task tracking	GitHub Projects board "public.safe — diploma" (used instead of YouTrack, as the form allows another tool): github.com/users/BekdauletBol/projects/1. 20 issues (#1–#15, #17–#21) with assignee, priority, due date, expected outcome, acceptance criteria and links to the implementing commits; 16 Done, 1 In progress, 3 Todo. Board screenshots are submitted separately (`Bolatov_TaskBoard.pdf`).
Traceability	Each issue lists the commit hashes that implement it (e.g. #10 → 8ec5122, 462c338, 62caa94); commit messages carry the milestone prefix (M0…M4). For an individual project, supervisor review of significant changes happens at the probation checkpoints.
Repository structure	`backend/publicsafe/` (api, ws, core, worker, db, eval), `backend/tests/` (unit, integration, security, cv, load), `frontend/src/` (pages, components, lib), `frontend/e2e/`, `docs/` (setup, operations, phone setup, evaluation, adr/, screenshots/, diagrams/, report/), `Makefile`, `docker-compose.yml`, `.env.example`, `.gitignore`

10.2 Milestones

Milestone	Scope	Issues	Status / date
M0	Repository scaffold, toolchain pinning, HTTPS dev setup	#1	Done, 10 Oct
M1	Database, migrations, admin auth, rate limiting, sessions/cameras/zones API	#2, #3	Done, 10 Oct
M2	PIN pairing, device credentials, camera and dashboard WebSockets, dashboard UI	#4–#6	Done, 10 Oct
M3	Frame ingest, mailbox scheduler, phone capture page, metrics, load test, E2E suite	#7–#9	Done, 10 Oct
M4	YOLOv8n detection, six rules, state machine, zone editor, overlays, events, CSV	#10–#12	Done, 10 Oct
—	Licence, docs cleanup, README for reviewers, tag prob-period-1	#17–#20	Done, 18 Oct
M5	Evaluation harness: datasets, annotation, metrics, rule comparison (E1–E3), latency benchmark (E5)	#13, #14, #15	In progress; due 7–21 Nov
—	Approbation report (PDF)	#21	Due 28 Nov
M6	Structured logging, API reference, demo checklist, 8-hour soak test, Android verification, E6/E7	to be created	Planned, Dec 2026 – Feb 2027

10.3 Risks and dependencies

#	Risk	Likelihood / impact	Mitigation
R1	Foot-point rules do not beat bbox overlap on public datasets (H1 fails)	Medium / high	Success criteria are pre-registered; a negative result is still reported with boundary analysis and the recommended rule is chosen by F1 — the system remains useful either way
R2	Laptop CPU/GPU throughput limits the number of phones (one worker, ~50 ms per frame)	High / medium	Latest-frame mailbox degrades gracefully (drops counted); YOLO11n or FP16 on CUDA behind the detector seam; target fps per camera configurable

#	Risk	Likelihood / impact	Mitigation
R3	Only one iPhone available; Android Chrome unverified	High / medium	Fake phones in tests and load test; borrow Android devices for E6; manual checklist in docs
R4	Weather, night and occlusion degrade detection outdoors	High / medium	E7 measures per-condition recall; confidence threshold per camera; hysteresis suppresses flicker; night variant of the detector as fallback [13]
R5	Dataset licences (MOT17 CC BY-NC-SA, WILDTRACK unstated) restrict publication	Medium / low	Non-commercial research use; confirm WILDTRACK terms with EPFL before publishing; own recordings with consent as third source
R6	AGPL-3.0 obligations of Ultralytics if the system is offered as a service	Low / medium	Repository itself is AGPL-3.0; permissive detector can replace the adapter (ADR 0009)
R7	Privacy / legal concerns about filming public space (Law of RK No. 94-V)	Medium / high	No video stored by default, no identity inference, privacy mode, streaming notice on the phone; signage at pilot sites
R8	Manual annotation of zone ground truth is slow and subjective	Medium / medium	Annotate every 5th frame, two annotators, Cohen's κ reported; annotation tool built into the CLI
R9	Time: M5 overlaps with other coursework; single developer	Medium / medium	Issues with due dates on the board; harness implemented incrementally (fetch → readers → metrics → report); supervisor checkpoints
R10	mkcert root certificate must be installed on every phone (manual on iOS)	High / low	Documented step with screenshots; served help page at /help/phone

Dependencies: Ultralytics weights download (once); MOT17 and WILDTRACK downloads; access to 2–3 phones for E6; supervisor availability for checkpoint reviews.

10.4 Next steps (to pre-defence 2)

1. Finish the evaluation harness (#13): dataset fetch with checksum, MOT17 and image-sequence readers, annotation tool, metrics with bootstrap and McNemar, report generation.
2. Draw and commit zone polygons for MOT17-02/04/09; annotate with two annotators; run E1–E3 and write `docs/EVALUATION.md` results.
3. Run E5 latency benchmark (#15) with 1/2/3/5 fake phones and E6 with real phones.
4. Verify Android Chrome; add F-ALERT-1 webhook for an external warning light.
5. Extend the literature list with truck blind-spot and VRU-detection sources; agree a pilot site (yard or depot) for own recordings.

11. Supervisor feedback

The supervisor, Nurgaliyeva Symbat Altybayevna, reviewed the student research work that precedes this project and reviews the project materials at the probation checkpoints. Her written review of the research work (reproduced below) assesses the problem analysis, the edge architecture, the implementation with YOLOv8/FastAPI/WebSockets and the foot-point + point-in-polygon method, and the experimental design; the degree of independence is rated as high.

ОТЗЫВ
научного руководителя
о степени самостоятельности выполненной работы

Научно-исследовательская работа «Edge-distributed pedestrian monitoring system using YOLOv8 and geometric spatial analysis for real-time urban danger zone detection», выполненная студентом Болатовым Бекдаулетом, демонстрирует высокий уровень самостоятельности на всех этапах научно-исследовательской деятельности — от постановки задачи до интерпретации полученных результатов.

В процессе выполнения работы студент самостоятельно провёл анализ современных подходов к обеспечению безопасности пешеходов и систем видеомониторинга, выявил ограничения централизованных CCTV-решений и обосновал необходимость разработки распределённой edge-архитектуры на базе мобильных устройств. На основе проведённого анализа была сформулирована цель исследования, определены гипотезы, задачи и выбрана методологическая база, включающая методы компьютерного зрения, пространственного анализа и экспериментальной валидации.

Студент самостоятельно разработал архитектуру интеллектуальной системы мониторинга, реализовал программную часть с использованием YOLOv8, FastAPI, WebSocket-взаимодействия и алгоритмов геометрической проверки (footpoint + point-in-polygon), а также выполнил экспериментальное тестирование прототипа в контролируемых условиях. В ходе работы были проведены измерения точности детекции, задержек обработки, сравнительный анализ с базовыми методами и интерпретация полученных метрик эффективности.

Особое внимание заслуживает способность автора к самостоятельному принятию инженерных решений: выбор вычислительной архитектуры, разработка алгоритма пространственной верификации, реализация механизма обработки видеопотока в реальном времени и организация экспериментального дизайна. Все этапы — от программной реализации до анализа результатов — выполнены студентом при минимальной консультационной поддержке научного руководителя.

Работа свидетельствует о сформированных навыках исследовательской деятельности, системного мышления, владения современными инструментами искусственного интеллекта и программной разработки, а также способности применять теоретические знания для решения прикладных задач в области интеллектуальных систем мониторинга.

Полученные результаты имеют практическую значимость и могут быть использованы при разработке интеллектуальных систем обеспечения безопасности городской инфраструктуры, а также в образовательных и исследовательских проектах в области компьютерного зрения и IoT-систем.

Степень самостоятельности выполнения научной работы оценивается как высокая.

Научный руководитель:
Нургалиева Сымбат Алтыбаевна
PhD, ассистент профессор школы компьютерной инженерии

Figure 11.1. Supervisor’s written review of the research work (original, Russian).

11.1 Feedback on key deliverables and resulting decisions

Deliverable	Supervisor's comment	Decision / action taken	Status
Research work: problem analysis and related work	Limitations of centralised CCTV and the need for a distributed edge architecture are well argued (review, para. 2)	Kept as the basis of Sections 1, 3 and 4 of this report; comparison extended from 4 approaches to 6 named systems and 8 criteria	Done
Research work: experimental design and metrics	Measurements of detection accuracy, latency and comparison with baselines are appropriate (review, para. 3)	Experiments re-registered with explicit success criteria (Section 9.4); evaluation moved from the lab-only data set to public datasets (MOT17, WILDTRACK) with bootstrap CIs and McNemar, because the lab ablation was found to be inconsistent with the stated rule	In progress (M5)
Research work: engineering decisions	Independent choice of computing architecture, spatial-verification algorithm and real-time stream handling is noted (review, para. 4)	Every such decision now recorded as an ADR with alternatives (Section 7.6); single shared queue replaced by per-camera mailbox (ADR 0006)	Done
Probation-period materials (this report, repository, board)	Review at the probation checkpoint after submission	Comments will be recorded in `docs/SUPERVISOR_FEEDBACK.md` and linked from the board	Pending

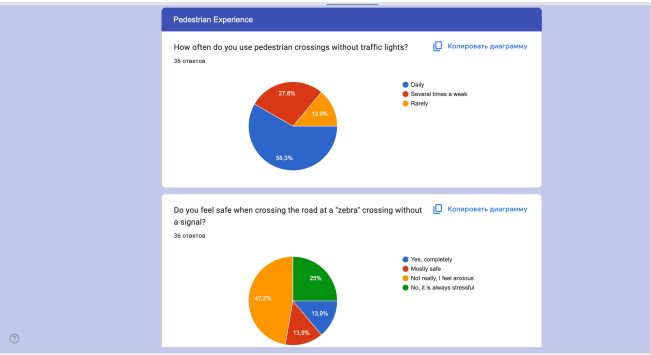
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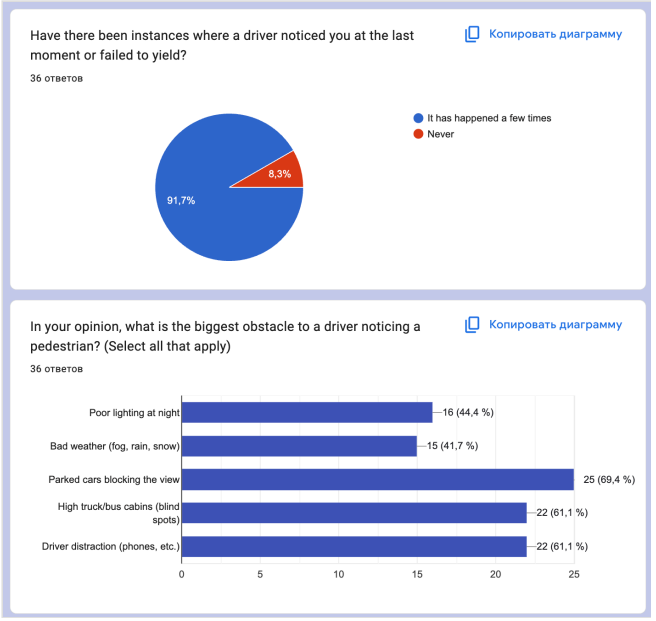
Vendor and project documentation used in Section 4: Mobileye Shield+ product page (ims.mobileye.com); Brigade Electronics, Sidescan Predict (brigade-electronics.com/detection-systems/sidescan-predict); Hikvision AcuSense overview; AXIS Object Analytics (axis.com/products/axis-object-analytics); Frigate NVR documentation, Zones (docs.frigate.video/configuration/zones); Ultralytics model documentation for YOLOv8 and YOLO11 (docs.ultralytics.com); WongKinYiu/yolov7 repository. Accessed October 2026.

Appendix A. Links to materials and survey screenshots

Material	Location (all at tag prob-period-1)
Repository / tagged version	github.com/BekdauletBol/public_final/tree/prob-period-1
README (overview, architecture, tests, status)	`README.md`
Implementation specification (requirements, API, CV pipeline, experiment plan)	`docs/IMPLEMENTATION_SPEC.md`
Architecture decision records	`docs/adr/0001…0009`
Diagrams — editable sources and exports	`docs/diagrams/*.puml`, `docs/diagrams/export/*.png|svg`
Evaluation plan, model checksum, FP16 measurement	`docs/EVALUATION.md`
Setup and operations	`docs/SETUP_MACOS.md`, `docs/PHONE_SETUP.md`, `docs/OPERATIONS.md`
Screenshots of the prototype	`docs/screenshots/m2/`, `m3/`, `m4/`
Automated tests	`backend/tests/`, `frontend/e2e/`
Task board	github.com/users/BekdauletBol/projects/1; screenshots in `Bolatov_TaskBoard.pdf`
Milestone commits	M0 794e733 · M1 48573c4 · M2 58fcc5e · M3 7211409 · M4 8ec5122 · history commits/prob-period-1
Research work (predecessor)	“Edge-distributed pedestrian monitoring system using YOLOv8 and geometric spatial analysis for real-time urban danger zone detection”, AITU, 2026 **[optional: add link]**
Survey (Google Forms)	Response summary screenshots below; **[optional: add form link]**



Survey response summary, pedestrian section (screenshot).



Survey response summary, obstacles and driver section (screenshot).